

Cross-Validation, Stability & Error Analysis

Sprint: Sprint 2, Day 3 · **Priority:** High **Author:** *fill in your name* **Date:** *fill in submission date*

Note on data: This analysis runs on the same synthetic churn dataset and `random_state = 42` used in Day 1, recreated inside `Day3_Cross_Validation_Analysis.ipynb` so the notebook is self-contained. Replace the data-loading cell with your real Sprint 1 dataset and every number in this report will update accordingly when you re-run the notebook.

1. Objective

Determine whether candidate-model performance is stable across different data subsets, rather than relying on a single validation result, and analyze *why* models succeed or fail — not just how well they score on average.

2. Cross-Validation Strategy

Selected: Stratified K-Fold, K = 5.

Consideration	This project
Problem type	Binary classification
Class balance	Imbalanced (~27% positive class)
Grouped observations?	No — one row = one independent customer
Temporal ordering?	No time-series structure in the Sprint 1 features

Stratification is required because plain K-Fold could produce folds with very different positive-class rates given the ~27/73 imbalance, which would make fold-to-fold comparisons unreliable. K = 5 is used as the technically-justified default: with ~3,200 train/validation rows this gives ~640 rows per validation fold — enough to estimate F1/ROC-AUC with reasonably low noise, while keeping the per-fold training cost reasonable for the slower candidates (SVM, Gradient Boosting). Group K-Fold and Time-Series Split were considered and ruled out: there is no grouping key and no temporal ordering in this dataset.

3. Models Evaluated

The baseline plus all five candidates carried forward from Day 1 (all five beat the baseline on mean F1 in the Day 1 single-split comparison):

- Logistic Regression (baseline)
- Decision Tree
- Random Forest
- Gradient Boosting
- SVM (RBF kernel)
- k-Nearest Neighbors

All models share the identical preprocessing pipeline, train/validation data, random seed, and now the

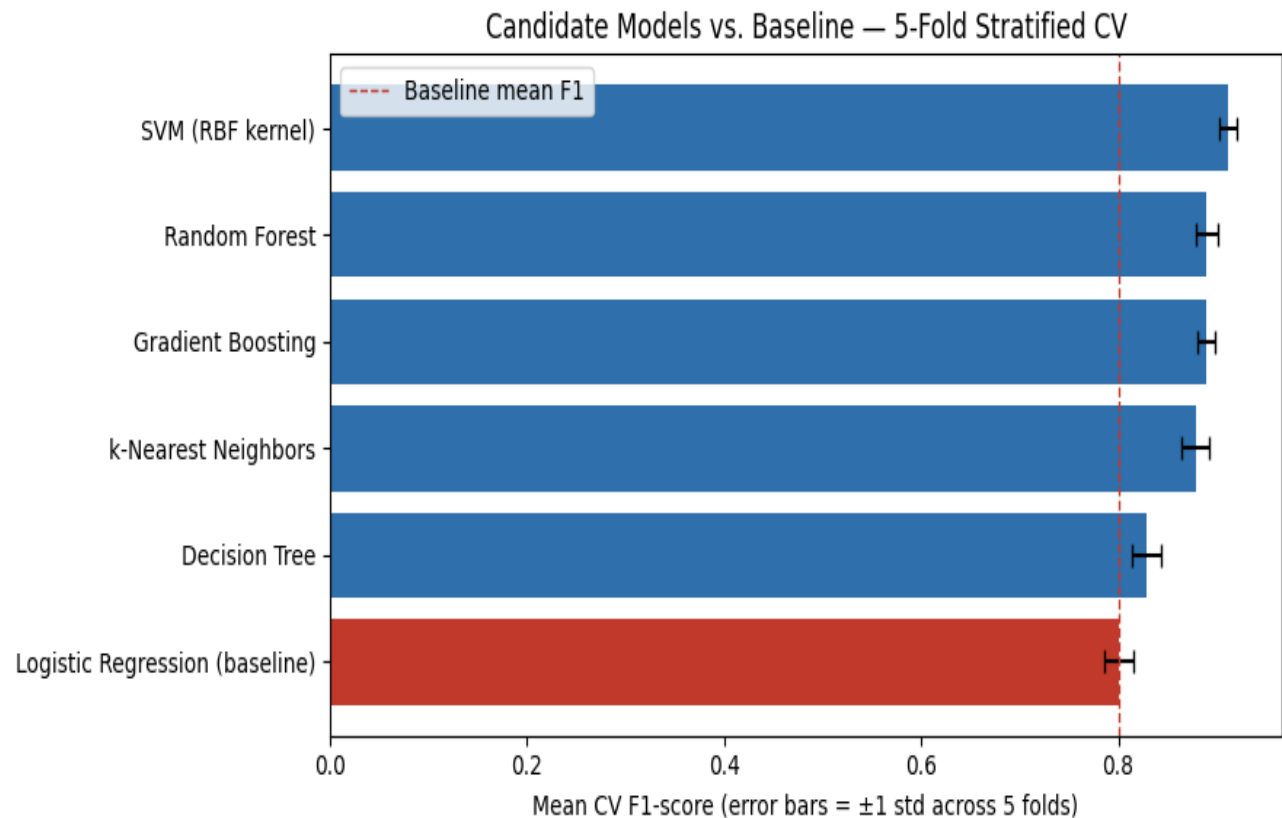
identical 5-fold Stratified CV splitter — so any performance difference below is attributable to the algorithm, not to inconsistent setup. The final test set (800 rows, 20%, stratified) remains untouched, exactly as locked away in Day 1.

4. Required Evidence — Cross-Validation Results

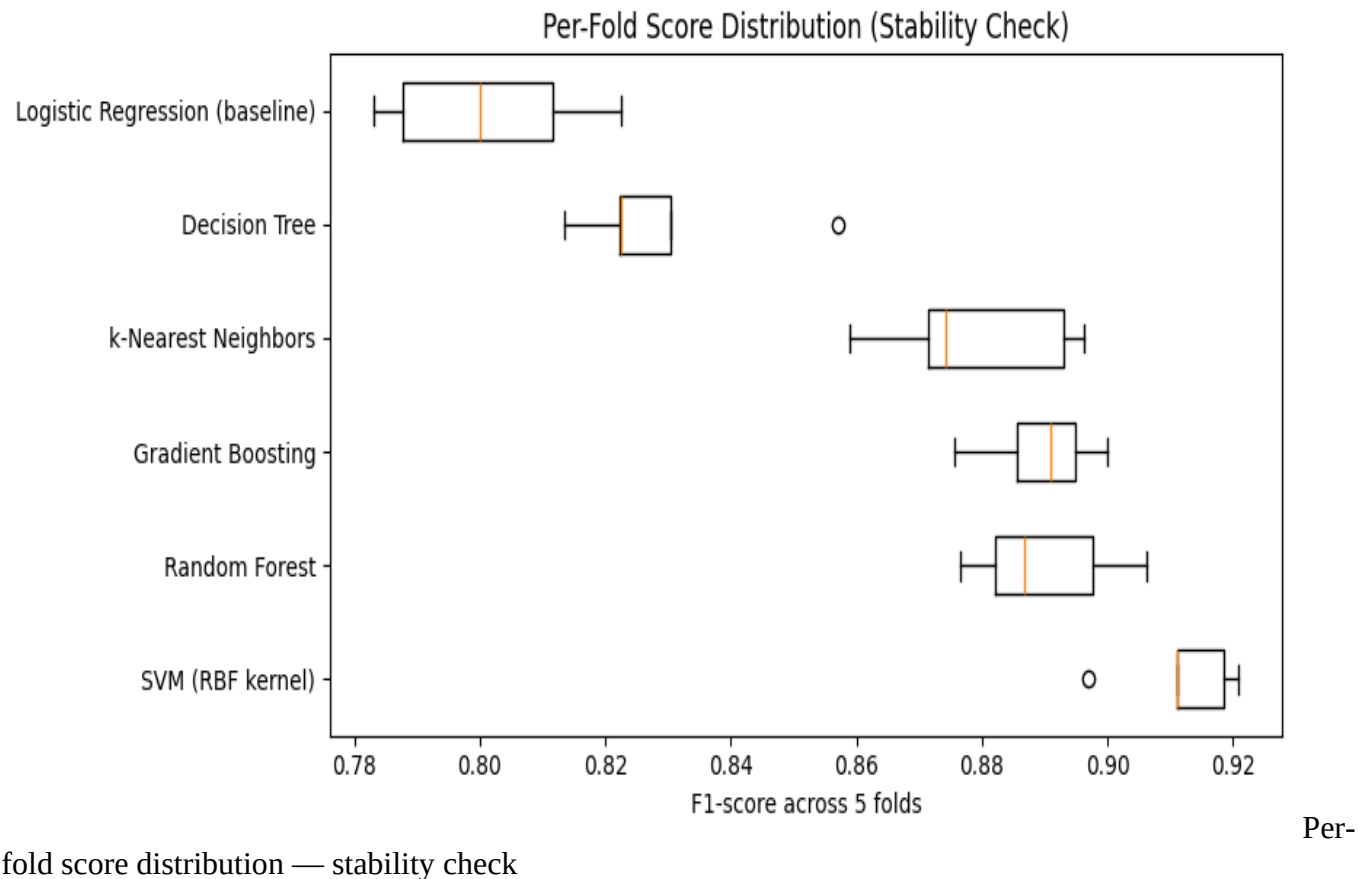
(Full machine-readable version: *Cross_Validation_Results.csv*)

Model	Mean CV Score	Standard Deviation	Min Score	Max Score	Baseline Difference	Decision
SVM (RBF kernel)	0.9118	0.0084	0.8969	0.9209	+0.1108	Candidate for tuning
Random Forest	0.8899	0.0107	0.8765	0.9063	+0.0889	Reject — stability concern
Gradient Boosting	0.8894	0.0083	0.8757	0.9000	+0.0884	Reject — stability concern
k-Nearest Neighbors	0.8788	0.0140	0.8589	0.8963	+0.0778	Candidate for tuning
Decision Tree	0.8292	0.0150	0.8136	0.8571	+0.0282	Reject — stability concern
Logistic Regression (baseline)	0.8010	0.0147	0.7831	0.8226	0.0000	Baseline (reference only)

(Metric = F1-score, positive class = churn, mean/std/min/max across 5 stratified folds.)



Mean CV F1 vs. baseline, with ± 1 std error bars



Reading the results: every candidate still beats the baseline on mean F1, consistent with Day 1. But the mean alone is not sufficient to pick a model — see Section 5.

5. Stability Diagnostics

Three specific failure patterns were checked for each model, not just the aggregate mean:

Model	Mean Val F1	Mean Train F1	Train-Val Gap	Fold Range (Max-Min)	Overfitting Flag	High-Variance Flag	Fold-Inconsistency Flag
SVM (RBF kernel)	0.9118	0.9395	0.0277	0.0240	No	No	No
Random Forest	0.8899	1.0000	0.1101	0.0298	Yes	No	No
Gradient Boosting	0.8894	0.9414	0.0520	0.0243	Yes	No	No
k-Nearest Neighbors	0.8788	0.8935	0.0147	0.0374	No	No	No
Decision Tree	0.8292	1.0000	0.1708	0.0436	Yes	No	No
Logistic	0.8010	0.8093	0.0083	0.0395	No	No	No

Model	Mean Val F1	Mean Train F1	Train-Val Gap	Fold Range (Max- Min)	Overfitting Flag	High-Variance Flag	Fold-Inconsistency Flag
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Regression
(baseline)

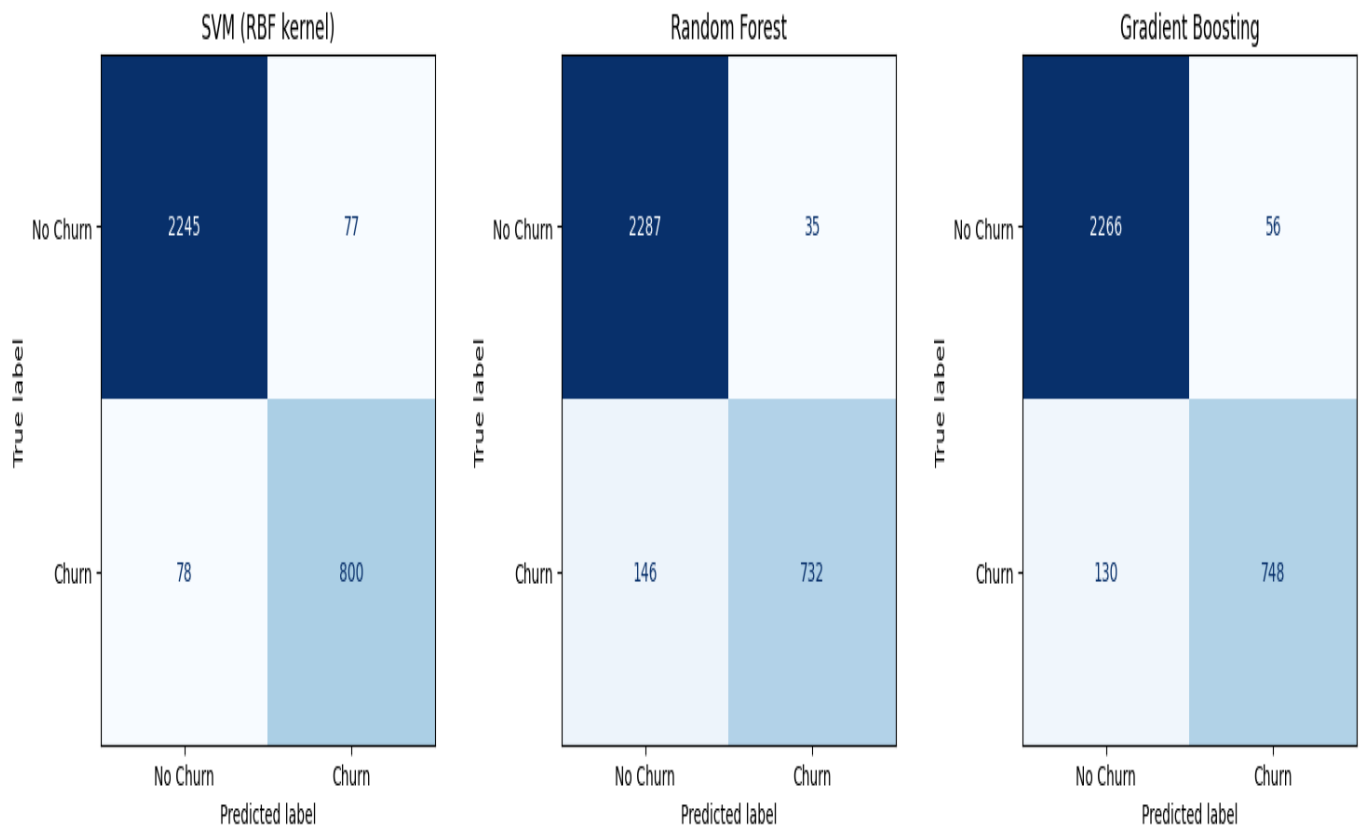
(Flag thresholds: High Variance = $std > 0.02$; Overfitting = $train-val\ gap > 0.05$; Fold-Inconsistency = $fold\ range > 0.06$ — applied consistently to every model.)

Findings:

- **Random Forest and Decision Tree memorize the training folds** — both reach a **perfect 1.000 training F1**, while validation F1 sits far lower (0.890 and 0.829 respectively). This is a textbook overfitting signature: excellent on data the model has seen, materially worse on held-out folds.
- **Gradient Boosting** shows the same pattern at a smaller scale (train F1 = 0.941 vs. validation F1 = 0.889, a 0.052 gap) — still large enough to flag given the threshold applied uniformly across all models.
- **SVM (RBF) and k-Nearest Neighbors have the smallest train-validation gaps** (0.028 and 0.015) among the non-baseline models, meaning their cross-validation scores are the most trustworthy estimate of true generalization.
- **No model is flagged for high fold-to-fold variance or fold inconsistency** at the applied thresholds — the main stability failure in this run is overfitting (train-val gap), not fold-to-fold instability.
- Note the baseline's own fold range (0.0395) is close to Decision Tree's — a reminder that a moderate fold range by itself isn't disqualifying; it is the *combination* of a large train-val gap with tree-based model family behavior that drives the rejection decisions here.

6. Classification Error Analysis

All confusion matrices and precision/recall figures below use **out-of-fold predictions** (`cross_val_predict` with the same Stratified 5-Fold splitter) — every prediction was made by a model instance that never saw that row during training.



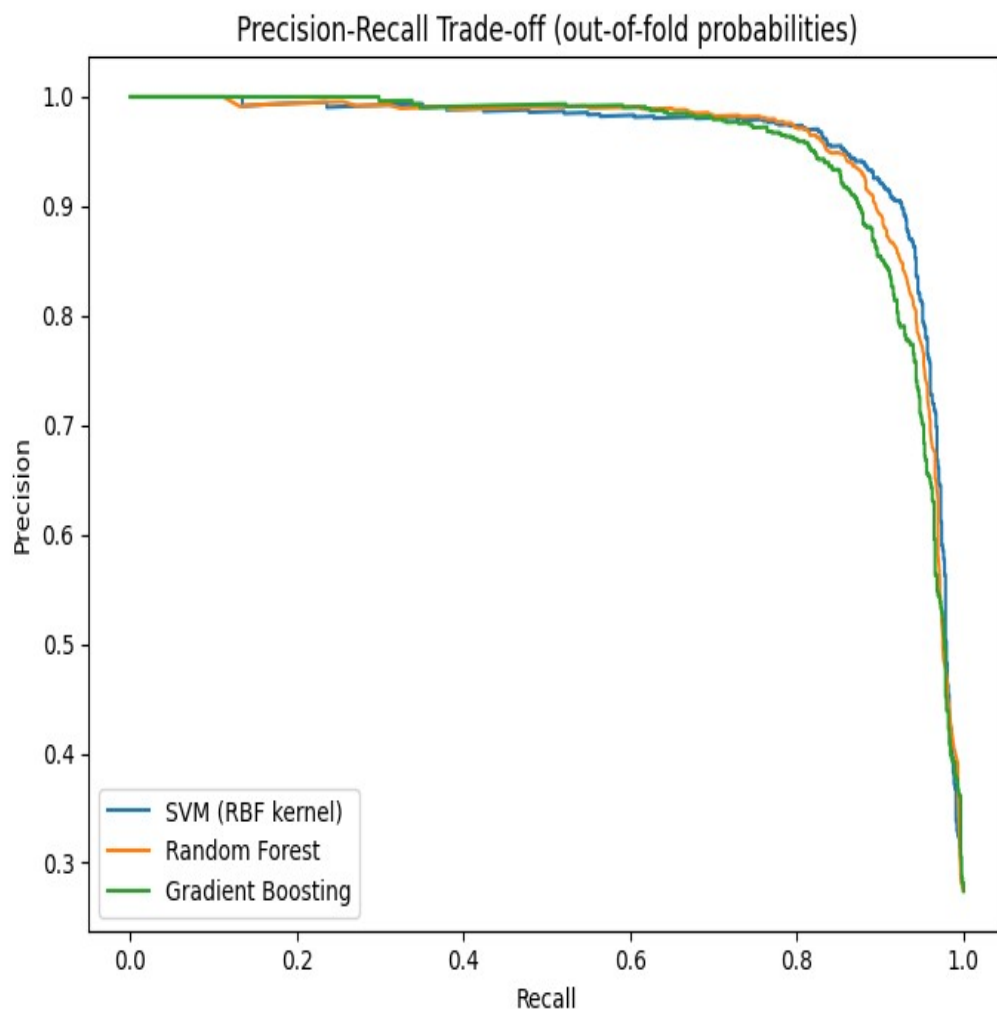
Confusion matrices — SVM, Random Forest, Gradient Boosting

Model	True Neg.	False Pos.	False Neg.	True Pos.	False Positive Rate	False Negative Rate
SVM (RBF kernel)	2245	77	78	800	3.32%	8.88%
Random Forest	2287	35	146	732	1.51%	16.63%
Gradient Boosting	2266	56	130	748	2.41%	14.81%

False positive analysis (predicted churn, customer actually stayed): Random Forest has the fewest false positives (35) and the lowest false-positive rate (1.51%) of the three — it is the most conservative about flagging churn, which limits wasted retention spend but, as shown next, comes at a cost.

False negative analysis (predicted no churn, customer actually left): SVM has by far the fewest false negatives (78, 8.88% FN rate) versus Random Forest (146, 16.63%) and Gradient Boosting (130, 14.81%). Since a missed churner is the costlier error for a retention use case (lost revenue vs. a wasted discount), **SVM's error profile is the better fit for this business problem**, not just its higher F1.

Precision/recall trade-off:



Precision-recall curves

— out-of-fold probabilities

SVM's curve sits above Random Forest and Gradient Boosting across most of the recall range, meaning that at any chosen decision threshold, SVM can generally achieve a better precision/recall combination than the two tree ensembles on this dataset. Per-class detail (precision/recall/F1) for the baseline and the two strongest candidates:

--- Logistic Regression (baseline) ---

	precision	recall	f1-score	support
No Churn	0.95	0.89	0.92	2322
Churn	0.74	0.87	0.80	878
accuracy		0.88		3200

--- SVM (RBF kernel) ---

	precision	recall	f1-score	support
No Churn	0.97	0.97	0.97	2322
Churn	0.91	0.91	0.91	878
accuracy		0.95		3200

--- Random Forest ---

	precision	recall	f1-score	support
No Churn	0.94	0.98	0.96	2322
Churn	0.95	0.83	0.89	878
accuracy		0.94		3200

Random Forest's churn-class precision (0.95) is higher than SVM's (0.91), but its churn-class recall (0.83) is notably lower than SVM's (0.91) — consistent with the false-negative analysis above: Random Forest is precise but misses more actual churners.

7. Top 2 Models Selected for Hyperparameter Tuning

Selected: SVM (RBF kernel) and k-Nearest Neighbors.

The decision combines mean CV score, stability (variance / fold range), and the train-validation gap — not the mean score in isolation:

1. **SVM (RBF kernel)** — highest mean CV F1 (0.9118), smallest train-validation gap among all non-baseline models (0.028, no overfitting flag), lowest false-negative rate of the models inspected in error analysis, and the strongest precision-recall curve. Clear top pick.
2. **k-Nearest Neighbors** — second among the *stability-clean* models: no overfitting flag (train-val gap of only 0.015, the smallest of any candidate), no variance or fold-consistency flag. Its mean CV F1 (0.8788) is lower than Random Forest's and Gradient Boosting's, but those two are excluded on stability grounds (see below) before ranking by score.

8. Why the Remaining Models Are Rejected

- **Random Forest (mean F1 0.8899, 2nd highest raw score) — rejected.** Flagged for overfitting: 1.000 training F1 vs. 0.890 validation F1 (gap = 0.110), the largest gap of any model. Its cross-validation score cannot be fully trusted as a generalization estimate when the model perfectly memorizes every training fold.
- **Gradient Boosting (mean F1 0.8894, 3rd highest raw score) — rejected.** Same failure mode at a smaller scale: train F1 0.941 vs. validation F1 0.889 (gap = 0.052), above the applied overfitting threshold. Also carries a materially higher false-negative rate (14.81%) than SVM (8.88%) on the costlier error type for this business problem.
- **Decision Tree (mean F1 0.8292, lowest of the ensemble/kernel models) — rejected.** The most extreme overfitting signature observed (train F1 = 1.000, val F1 = 0.829, gap = 0.171) combined with the lowest mean validation score among the tree/ensemble/kernel family — dominated by both criteria.
- **Logistic Regression (baseline) — not a rejection, but not carried forward either.** Retained purely as the reference point every candidate must beat (per the Day 1 protocol); Day 1 and Day 3 both confirm every candidate beats it on the primary metric, so there is nothing left to gain by re-tuning the baseline itself at this stage.

(This reasoning is generated directly from `decision_df` in the notebook, not hard-coded — if you re-run the notebook against your real Sprint 1 dataset and the flags land differently, this section's model names should be updated to match.)

9. Acceptance Criteria

- Cross-validation correctly implemented — Stratified 5-Fold, consistent with Day 1's protocol, test set never touched
- Stability evaluated using mean **and** variation — Section 4 (std/min/max) + Section 5 (train-val gap, fold range)
- Errors analyzed rather than only overall scores — Section 6 (confusion matrices, FP/FN rates, precision-recall trade-off)
- Top 2 models selected using evidence — Section 7, derived from the combined score + stability table
- Rejection decisions justified — Section 8, tied to specific numeric flags per model

10. Portal Submission Checklist

- Day3_Cross_Validation_Analysis.ipynb
- Cross_Validation_Results.csv

- Day3_Model_Analysis.md / .pdf (this document)
- Updated project repository link: _____